

RF-One: Is It Worth Building?

Response on Desire and Goal, and on the commercial case

From Shelbi Fox | Date 17 August 2026 | Follows RF-One Model Review, first pass

You asked the hardest question there is, so I have put the answer at the top rather than making you read to the end for it.

Short answer: yes, the paradigm is worth pursuing. But the way you have described it is not yet what makes it different, and the version you are currently pitching is a fight you would lose.

The white space is real. It is just narrower, and further underneath the surface, than "we start from the goal instead of the data." I will show you what I mean.

First, where you are right and I was wrong

On Desire and Goal, you are correct and I withdraw the objection.

My criticism was that Goal-First is impossible under the Core's own definition, because Goal.md requires a Process to exist before a Goal does, while every strategy paper starts from what the owner wants. Your answer resolves that completely. If client ambitions enter the system as **Desires**, and the work of the AI is to convert them into Goals by analysing resources, conditions and constraints, then the Core and the strategy papers are perfectly consistent. Goal-last is the correct sequence. I read an ordinary-language use of "goal" in the strategy papers as the canonical one, and that was my error.

I will go further than you did, because I think you are underselling it. **The Desire to Goal transition is the best idea in the project.** A system that converts vague ambitions into structured, resourced, executable plans is a genuinely novel framing, and unlike most of the model it is authentically domain independent. That is the thing that justifies having a Core at all. You are treating it as a definitional footnote. It is closer to being the product.

But the clarification exposes a real gap

Your own sentence describes the mechanism: RF-One takes a Desire, analyses **Resources, Conditions and Constraints**, applies knowledge, and constructs a **Process**. You close by saying Desires become Goals through **Processes, Decisions and Actions**.

Six concepts. Here is where they stand in the Core today:

Concept	Status in the current Core
Process	Defined, and defined well
Desire	Not defined. Appears once, inside Goal.md, referring to nothing. Full document exists only in Old/
Resource	Not defined. Appears only as a role an Entity may assume
Condition	Not defined
Constraint	Not defined. Empty stub in Old/

Concept	Status in the current Core
Decision	Explicitly excluded - Entity.md lists it as not an Entity
Action	Not defined

So this is not a wording inconsistency. The transition you call the fundamental job of the AI has no vocabulary in the model. I do not think that is a flaw so much as the next piece of work, and probably the most valuable piece available right now. Old/Desire.md already had identity, persistence and a full status lifecycle. You solved part of this once and lost it in the restructure.

Now the real question: is it worth building?

I went and checked the market rather than giving you an opinion. Everything below is dated and attributed so you can verify it.

Where I think you are wrong about the competition

Your framing is that incumbents collect data and build dashboards, and RF-One starts from the objective. That framing is already taken.

5-Out is an AI restaurant forecasting company whose homepage headline is literally "**Action, not Insights.**" It predicts sales and then recommends labour schedules, purchasing and prep. That is your pitch, in market, today.

Toast expanded Toast IQ in May and June 2026 into an AI assistant that answers plain-language questions and takes actions inside the platform, including a marketing agent that builds audiences and runs campaigns. **Clover** went a different route in May 2026, partnering with Tabit to launch Clover Reserve for fine dining with AI-driven operational intelligence.

If you walk into a room and say "we recommend actions instead of showing dashboards," the operator has heard it, and the people who said it first have far more money than we do.

Where I think you are right, and it is better evidenced than your version of it

The interesting finding is that restaurant AI is broadly failing, and the reason given is your thesis.

- Taco Bell scaled drive-thru voice AI and had to rethink it after inconsistent performance. McDonald's ended its voice AI partnership with IBM in 2024.
- The diagnosis across the trade press is that AI is being layered onto disconnected systems rather than integrated into unified workflows.
- **James Schonzeit, Head of Food and Beverage at Square**, said publicly at the National Restaurant Association Show in May 2026 that AI is limited by the data it can reach, that incomplete visibility prevents good decisions, and that some technology firms isolate data and block operator insight.
- **Oliver Ostertag, President of Growth and AI at Par Technology**, said in July 2026 that AI works when it sits on deep data that understands inventory and labour and ties both back to in-store sales.

Two executives at incumbent platforms are describing your architecture as the missing piece. **That is a much stronger argument than "they only do dashboards," and it is not coming from us.**

Worth noting alongside it: Crunchtime, Restaurant365 and MarginEdge all added AI in 2026, but as modular features bolted onto stacks designed decades ago. Architecture is your differentiator. Features are not.

The counter-argument you should have an answer for

Kevin Bryla, CMO of SpotOn, at the same event: AI functions best as a decision-support tool rather than a comprehensive solution, and it addresses limited problems rather than universal business issues. That is a direct rebuttal of the RF-One thesis from someone selling into the same buyer. It deserves a real answer, not a dismissal.

And the demand signal is weak right now

The National Restaurant Association's State of the Restaurant Industry 2026, published in February, puts AI adoption at **26% of operators** - and it is concentrated in marketing (19% of full-service, 15% of limited-service) and administration (10%). Customer ordering is 6%. Decision optimisation does not register.

That cuts both ways. It is genuine white space. It also means nobody is currently buying what we would be selling, so we would be creating the category as well as the product.

Two questions I will be asked in every conversation

I would rather have your answers before I am in the room than improvise them. Neither of these is a criticism of the model. Both are cheaper to design around now than to discover after there is code.

1. What does it cost us to serve one restaurant?

Brendan Sweeney, CEO of Popmenu, at the NRA Show in May 2026, made the point that AI token costs can exceed the salaries of the roles they replace, and that rising energy costs make AI services unlikely to get cheaper.

RF-One as described is the most compute-heavy design in the category: continuous observation, simulation, causal reasoning, digital twin. Meanwhile mid-tier restaurant POS software runs roughly **\$60 to \$200 per month per location**, and software is typically only a small fraction of an operator's total technology bill.

So: what is our price point, and what does an account cost us to run? If the answer is that we do not know yet, that is fine at this stage - but it should shape the architecture, and you are at exactly the right moment to let it. Not everything needs the most expensive model on every cycle. Cheap monitoring with expensive reasoning on triggers is a design decision, not a later optimisation.

2. What happens when the prospect is on Toast?

Toast's standard self-service API tier is **read-only** and requires the restaurant to hold a qualifying subscription. Deeper partner access runs through a formal multi-stage certification programme. Commercial terms need confirming directly with Toast - I am not going to quote you a number I cannot verify.

This is not a wall. MarginEdge, Restaurant365 and 5-Out all integrate with Toast successfully. It is a process with a clock on it. If Toast integration is on the roadmap at all, it may be worth starting the partner application in parallel with the modelling rather than after it, because certification time is time we cannot compress later.

I would also gently point out that the Clover folder in the repository currently contains eleven files, all of them empty. The dependency we are most exposed to is the one we have explored least.

The hardest problem in your pitch

Your example is: replace this dish with that one and expect roughly \$5,000 more per week.

That specific claim is the hardest thing in the entire concept, and I want to be honest about why. It is a causal question, not a data question. To make it you need substitution effects, cross-price elasticity, capacity constraints and margin structure - and you need to detect the result against the noise of a real restaurant.

A restaurant doing around \$58,000 a week routinely swings ten to twenty percent week to week on weather, events and seasonality. A \$5,000 effect is smaller than the normal noise. Detecting it by observation alone would take something like thirty weeks of stable conditions, and restaurants are never stable for thirty weeks. More model sophistication does not fix this. It is a sample size problem.

Three things follow, and I think they are all constructive:

- 1 The Experiment Engine is not function number six. It is the foundation.** Controlled tests are the only rigorous route to causal claims at single-restaurant scale. I would promote it to the centre of the architecture.
- 2 Your second example is far better than your first.** The pasta station capping Friday occupancy at 82%, worth around \$38,000 a year, is measurable directly from ticket times and throughput. No counterfactual required. Constraint Intelligence is the defensible wedge. Menu engineering is the victory lap. I would lead with the bottleneck story.
- 3 Decompose the claim.** Do not predict a revenue effect. Predict margin, forecast volume and substitution rate separately. Each is measurable on its own.

There is also an opportunity hiding in this. Every ROI figure I found in this market - "15 to 25% margin improvement," "8 to 12% revenue growth" - traces back to vendor marketing. I could not find a single neutral benchmark for restaurant AI recommendation accuracy. **In a market full of unverifiable claims, publishing honest calibration and real hit rates would itself be a differentiator.** That is exactly what an Experiment Engine would produce.

What I would do in the next ninety days

You are asking whether RF-One is worth building as though it were a question of judgment. I think it is a question of evidence, and you already own the laboratory.

Take one real Desire from Winter Park. Let RF-One convert it into a Goal - resources, conditions, constraints, process. Even done by hand, by you, on paper. Produce **one specific, dated, falsifiable prediction with a dollar value.** Write it down in advance. Execute it. See what happens.

If the paradigm can produce one correct, non-obvious, economically material recommendation on a real restaurant, that is worth more than another hundred documents, and it is the single most persuasive thing we could show an investor or a customer. If it cannot, we have learned that for the price of a fortnight.

Mount Dora makes the timing better than it will ever be again. A second location exercises multi-location, per-location menus and per-location pricing all at once, on a real deadline. I would run it through the model on paper before it opens.

Questions I think you should answer for yourself

- 1** Venture-scale company or highly profitable tool? Almost every other decision depends on this one.
- 2** Who is the paying customer - independents, or five-to-fifty location groups? The strategy papers aim the design test at the least sophisticated operator, who is the right proof but possibly the wrong market.

- 3 What is the plan if Toast and Clover restrict data access?
- 4 What would falsify the paradigm? If nothing would, it is a belief rather than a thesis.
- 5 What would tell you the Core is ready to build on? I asked this in the first review and it is still the question I would most like answered.

Where I land

Yes, I think it is worth building - with two changes to how you are describing it.

Stop selling the inversion and start selling the architecture. "We start from the goal, not the data" is a crowded claim. "Restaurant AI is failing because it sits on disconnected systems, and we modelled the business before we wrote the software" is defensible, it is supported by the incumbents' own executives, and it is true of you in a way it is not true of them.

And lead with the claims you can prove. Bottlenecks and cost leakage are observable. Dish substitution revenue is a research problem. Win on the first, then earn the right to the second.

The Desire to Goal engine is the genuinely original idea here, and right now it is the least documented thing in the repository. If you want my recommendation on where the next month goes, it is there - not in Purchasing, which we now know how to fix.

Happy to keep going on any of this. If we are heading toward taking it to market, we should also have a short conversation about how that works between us, so we are both clear before I start making introductions rather than after.

Shelbi Fox | 17 August 2026 | Working response, prepared for discussion

Sources

Market claims above are drawn from the following, checked 17 August 2026:

- National Restaurant Association, *State of the Restaurant Industry 2026*, published February 2026 - AI adoption figures.
- Restaurant Dive, May 2026 - National Restaurant Association Show panel remarks by Brendan Sweeney (Popmenu), Kelly Esten (Toast), James Schonzeit (Square), Kevin Bryla (SpotOn).
- Restaurant Dive, July 2026 - Oliver Ostertag (Par Technology) on where restaurant AI works.
- Restaurant Technology News, June 2026 - Toast IQ expansion and Spring 2026 release.
- Fiserv investor news release, May 2026 - Clover Reserve powered by Tabit.
- 5-Out company website, 5out.io, checked 17 August 2026 - positioning and product claims.
- Toast developer documentation, doc.toasttab.com, checked 17 August 2026 - API access tiers and requirements.
- Restaurant POS and software cost figures are from 2026 industry comparison sources and should be treated as indicative rather than precise.